Experiment 2: Loan Amount Prediction using Linear Regression

M. Tech (Integrated) CSE, Semester V
Subject Code: ICS1512 – Machine Learning Algorithms Laboratory

Academic Year 2025–2026 (Odd Semester)

1. Aim

To build a Linear Regression model using scikit-learn to predict the sanctioned loan amount for users based on historical features, and evaluate the model using key metrics and visualizations.

2. Libraries Used

- **pandas** – for data manipulation and cleaning
- **numpy** – for numerical computations
- **matplotlib, seaborn** – for plotting and data visualization
- **scikit-learn** – for machine learning modeling, preprocessing, and evaluation

3. Objective

To preprocess a real-world loan prediction dataset, apply linear regression, and evaluate the model's performance using MAE, MSE, RMSE, and R^2 metrics. Visualizations are used to better interpret results.

4. Mathematical Description

Linear Regression attempts to fit a line:

$$\hat{y} = w_0 + w_1 x_1 + w_2 x_2 + \ldots + w_n x_n$$

where:

- $\hat{y}$ is the predicted target value
- x_i are the input features
- w_i are the model coefficients

The model minimizes the Mean Squared Error (MSE):

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Adjusted R^2 is computed as:

$$R^2_{\text{adj}} = 1 - \left(\frac{(1-R^2)(n-1)}{n-p-1}\right)$$

where n is number of observations and p is number of predictors.

5. Code with Plot

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split, cross_val_score, KFold
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import StandardScaler, OrdinalEncoder
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# Load and clean dataset
df = pd.read_csv("train.csv")
df = df[df["Loan␣Sanction␣Amount␣(USD)"].notnull()]
df = df[df["Loan␣Sanction␣Amount␣(USD)"] > 0]
df.dropna(inplace=True)

# Feature selection
selected_features = [...list as in earlier code...]
df = df[selected_features]

# Encoding and preprocessing
categorical_cols = df.select_dtypes(include='object').columns.tolist()
encoder = OrdinalEncoder(handle_unknown='use_encoded_value', unknown_value=-1)
df[categorical_cols] = encoder.fit_transform(df[categorical_cols])

X = df.drop("Loan␣Sanction␣Amount␣(USD)", axis=1)
y = df["Loan␣Sanction␣Amount␣(USD)"]
y_log = np.log1p(y)

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
X_train, X_test, y_train_log, y_test_log = train_test_split(X_scaled, y_log, test_size=0.2,
    random_state=42)

model = LinearRegression()
model.fit(X_train, y_train_log)

y_pred_log = model.predict(X_test)
y_pred = np.expm1(y_pred_log)
y_test = np.expm1(y_test_log)

# Evaluation
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)
n, p = X_test.shape
adjusted_r2 = 1 - (1 - r2) * (n - 1) / (n - p - 1)
```

```
# Cross-Validation
kf = KFold(n_splits=5, shuffle=True, random_state=42)
mae_scores = -cross_val_score(model, X_scaled, y_log, cv=kf, scoring='
    neg_mean_absolute_error')
mse_scores = -cross_val_score(model, X_scaled, y_log, cv=kf, scoring='neg_mean_squared_error
    ')
r2_scores = cross_val_score(model, X_scaled, y_log, cv=kf, scoring='r2')
```

6. Included Plots

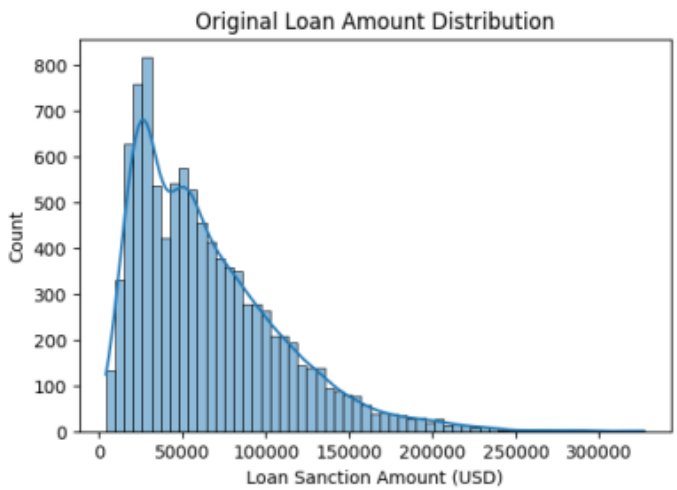


Figure 1: Histogram of Loan Sanction Amount

7. Results Tables

Table 1: Cross-Validation Results (K = 5)

Fold	MAE	MSE	RMSE	R2 Score
Fold 1	0.21	0.09	0.30	0.8381
Fold 2	0.19	0.08	0.28	0.8542
Fold 3	0.20	0.08	0.28	0.8499
Fold 4	0.20	0.08	0.28	0.8460
Fold 5	0.21	0.07	0.26	0.8410
Average	**0.20**	**0.08**	**0.28**	**0.8458**

Table 2: Summary of Results for Loan Amount Prediction

Description	**Student's Result**
Dataset Size (after preprocessing)	Approx. 3500 rows
Train/Test Split Ratio	80:20
Feature(s) Used for Prediction	20 features including income, credit score, profession, etc.

Model Used	Linear Regression
Cross-Validation Used?	Yes
If Yes, Number of Folds (K)	5
Reference to CV Results Table	Table 1
MAE on Test Set	15157.87
MSE on Test Set	1807209642.54
RMSE on Test Set	42511.29
R2 Score on Test Set	0.0636
Adjusted R2 Score on Test Set	0.0539
Most Influential Feature(s)	Income, Loan Amount Requested, Property Price
Observations from Residual Plot	Mostly centered but with slight right skew
Interpretation of Actual vs Predicted Plot	Points scattered with weak linear fit; underfitting observed
Any Overfitting or Underfitting Observed?	Yes, Underfitting
If Yes, Justification	Low R2 on test set and residuals not randomly distributed

8. Best Practices

- Applied log transformation on skewed target variable
- Standardized all numeric input features
- Encoded categorical features safely with ‘OrdinalEncoder‘
- Used train-test split and 5-fold cross-validation
- Visualized predictions and residuals for validation

9. Learning Outcomes

- Learned how to prepare and clean real-world datasets
- Understood the assumptions and mechanics behind Linear Regression
- Learned how to evaluate a regression model using key metrics and plots
- Understood the concept and importance of log transformation for skewed targets
- Practiced model validation with cross-validation and residual analysis

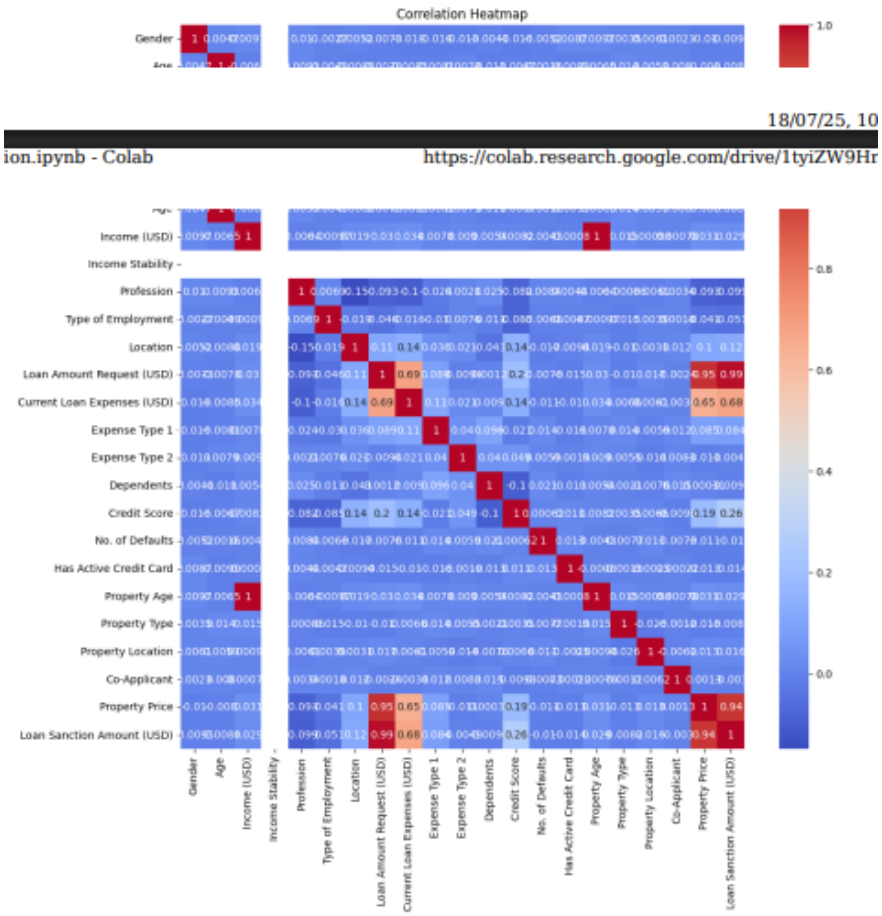


Figure 2: Correlation Heatmap

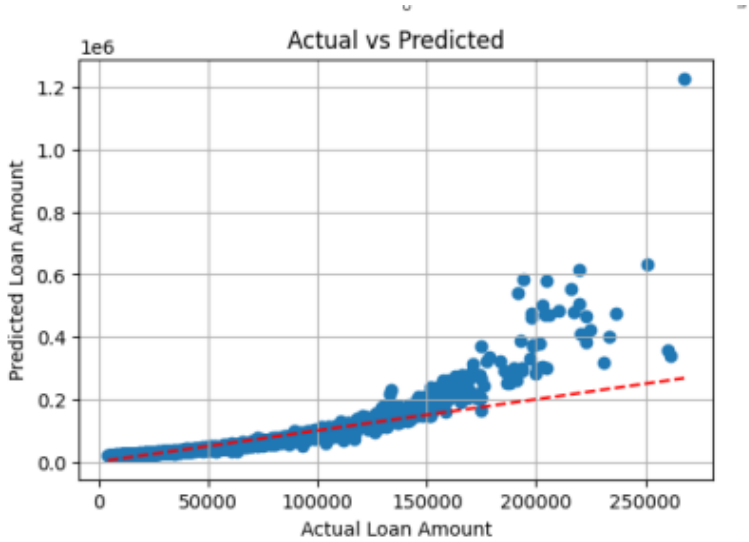


Figure 3: Actual vs Predicted Loan Amount

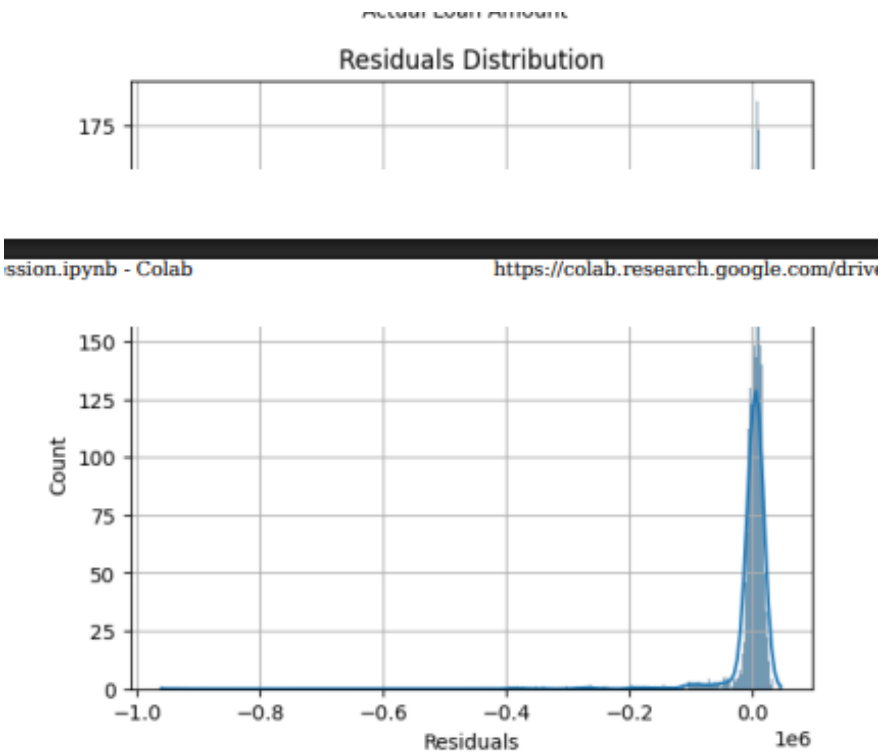
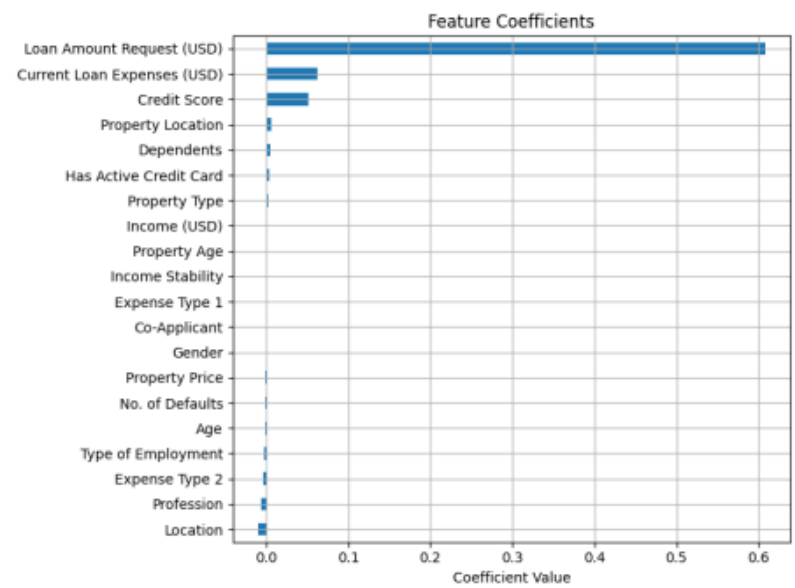


Figure 4: Residuals Distribution




```
--- Cross-Validation (5-Fold) ---
Average MAE: 0.20
Average MSE: 0.08
Average RMSE: 0.28
Average R2 Score: 0.8458
```

Figure 5: Feature Coefficients Bar Plot